

Comparative Analysis of Classical and Deep Learning Architectures for Time Series Forecasting *Group-7*

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Abstract

Time series forecasting relies on two fundamentally distinct paradigms: classical statistical methods, grounded in linear assumptions, and modern deep learning architectures, capable of modeling complex, non-linear dependencies. This paper presents a comparative analysis of these methodologies, specifically evaluating the performance of Box-Jenkins models (SARIMA, ARIMAX) against Long Short-Term Memory (LSTM) networks. Through empirical experimentation on practical datasets, we demonstrate the inherent limitations of utilizing these models in isolation across varying degrees of multivariate complexity and non-stationarity. Also we try to introduce some ensemble methodology will fuse both paradigms.

1. INTRODUCTION

Time series forecasting—predicting future values from sequences of historically observed data—is a foundational problem in applied machine learning. It underpins critical infrastructure across diverse domains: energy grid management, retail inventory planning, transportation scheduling, and financial risk modeling all depend on accurate, forward-looking predictions.

Historically, the dominant approach to this problem was rooted in classical statistical methodologies. The Box-Jenkins family of models—most notably ARIMA and its seasonal extension, SARIMA—served as the standard benchmark for univariate forecasting throughout the late 20th century. These models offered principled, interpretable tools grounded in rigorous probability theory, while extensions such as ARIMAX enabled the incorporation of exogenous predictor variables. However, these classical models are based on certain assumptions: they presuppose a linear relationship between historical and future values, and they generally assume that error residuals follow a normal distribution. While these models perform well on stationary, seasonal data, they do have many limitations.

The deep learning revolution of the 2010s fundamentally disrupted this forecasting landscape. Recurrent neural networks (RNNs), and specifically Long Short-Term Memory (LSTM) architectures, demonstrated a profound capacity to model complex, non-linear temporal dynamics directly from raw data, circumventing the need for extensive manual feature engineering. More recently, Transformer architectures—originally developed for natural language processing—have been adapted for time series analysis, offering improved handling of long-range temporal dependencies via self-attention mechanisms.

In the first half of the paper we attempt to do a comparative analysis between classical and deep learning models, we first take a less contextual dataset and then we train on dataset with higher complexity and non-linearity.

2. RESEARCH QUESTIONS

(RQ1) Under what specific temporal conditions do classical statistical models outperform LSTM-based deep learning architectures, and vice versa?

(RQ2) How effectively can linear models (ARIMAX) leverage multivariate exogenous signals compared to the non-linear processing capabilities of an LSTM architecture?

(RQ3) To what extent can hybrid architectural extensions—such as classical-neural ensembles or CNN-LSTM networks—yield further predictive improvements on multivariate demand and highly seasonal forecasting tasks?

2.1. BRIEFINGS ABOUT THE DATASETS USED

2.1.1 UCI Bike Sharing Dataset (Hourly) The first dataset is the UCI Bike Sharing Dataset, which records hourly bicycle rental counts in Washington D.C. from January 2011 to December 2012 — approximately 17,379 hourly observations. The target variable is the total rental count (cnt), which combines casual and registered users. The dataset includes a rich set of exogenous variables that capture contextual information about each hour.

Tabela 1 – UCI Bike Sharing Dataset

Feature	Description	Type
holiday	Whether the day is a public holiday	Binary
workingday	Whether the day is a working day	Binary
weathersit	Weather situation (1=clear, 4=heavy rain)	Ordinal (1–4)
temp	Normalised temperature in Celsius	Continuous
atemp	Normalised “feels like” temperature	Continuous
hum	Normalised humidity	Continuous
windspeed	Normalised wind speed	Continuous

The hourly rental series exhibits strong intraday seasonality (morning and evening commute peaks), weekly seasonality (weekday vs. weekend usage patterns), and long-term trend growth over the two-year observation window. These characteristics make it well-suited for testing both seasonal classical models and deep learning architectures capable of learning multi-scale temporal patterns.

2.1.2 Champagne Sales Dataset (Monthly) The second dataset records monthly champagne sales for Perrin Frères across a multi-year observation window. It is a univariate, monthly-frequency series with strong annual seasonality — sales peak sharply in December (holiday season) and trough in January-February. This makes it a canonical dataset for evaluating SARIMA’s seasonal differencing and seasonal MA components. Unlike the bike sharing series, this dataset contains no exogenous variables, making it exclusively suitable for univariate modelling. A 24-month (2-year) hold-out test set is used for evaluation, consistent with the convention for monthly forecasting benchmarks

3. METHODOLOGY

3.1 Preprocessing Both pipelines follow a common preprocessing strategy, with dataset-specific adaptations:

Datetime indexing: Both datasets are indexed by a datetime column. For the bike sharing data, the separate date (dteday) and hour (hr) columns are concatenated and parsed into a single datetime index. For the champagne dataset, the monthly string column is parsed directly. Chronological sorting is enforced after indexing.

Train/test splitting: A strict chronological split is used in all cases — the test set always consists of the most recent observations, with no shuffling. This is a non-negotiable requirement for time series evaluation: random splitting would constitute look-ahead leakage, since the model could observe future data during training. The bike sharing models use a 168-hour (7-day) test horizon; the champagne model uses a 24-month horizon.

Scaling (LSTM only): Neural networks are sensitive to input scale, particularly with the sigmoid and tanh activations used inside LSTM gates. All features are scaled to $[0, 1]$ using MinMaxScaler. Critically, the scaler is fitted exclusively on training data and then applied to the test set — fitting on the full dataset would constitute a subtle form of data leakage by allowing the model to “see” the range of future values during scaling.

Sequence construction (LSTM only): The LSTM requires input in the form of 3D tensors of shape (samples, timesteps, features). A sliding window of length 24 hours is used to construct input sequences: for each hour i , the model receives the 24 preceding hours of all features as input and predicts the rental count

at hour $i+1$. This window length was chosen to capture a full daily cycle of usage patterns.

3.1. MODEL ARCHITECTURE

Here is a brief discussion on implementation of models and their architecture

SARIMA (Seasonal ARIMA) extends the ARIMA framework with additional seasonal autoregressive and moving average terms. The model is parameterised as $\text{SARIMA}(p, d, q)(P, D, Q, s)$, where (p, d, q) govern the non-seasonal AR order, differencing degree, and MA order respectively, and (P, D, Q, s) govern the corresponding seasonal components with period s . For the champagne dataset, we use order $(1,1,1)$ with seasonal order $(0,1,1,12)$, reflecting one round of seasonal differencing (to remove the annual seasonality), one regular differencing (to remove trend), and first-order non-seasonal and seasonal MA terms. The seasonal period $s=12$ corresponds to the annual cycle in monthly data.

ARIMAX extends ARIMA by incorporating a vector of exogenous predictor variables (X) alongside the autoregressive and moving average terms. The exogenous variables enter the model as a linear regression component on top of the ARIMA structure. For the bike sharing dataset, the seven contextual features (weather, temperature, humidity, holiday flags, wind speed) serve as exogenous inputs. The model uses order $(2,1,2)$ — two autoregressive lags, one round of differencing, and two moving average terms. One round of differencing is applied to achieve approximate stationarity in the rental count series. As noted, the model is fitted on a 720-hour (30-day) subset due to the $O(n^2)$ computational cost of SARIMAX on long series.

An important practical note: ARIMAX requires the exogenous variables to be available for the test period at forecast time. This means it cannot be used for pure out-of-sample forecasting without knowing future weather and calendar values — a realistic constraint, since weather forecasts are typically available at 24-48 hour horizons with reasonable accuracy.

And lastly, discussing about LSTM, **The LSTM model** is a sequential neural network with the following architecture:

The model is compiled with the Adam optimiser and mean squared error loss, trained for 20 epochs with a batch size of 64. A 10 percent validation split from the training data is used for early stopping monitoring. Unlike the ARIMAX model, the LSTM trains on the full 17,379-row history, giving it substantially more signal to learn from.

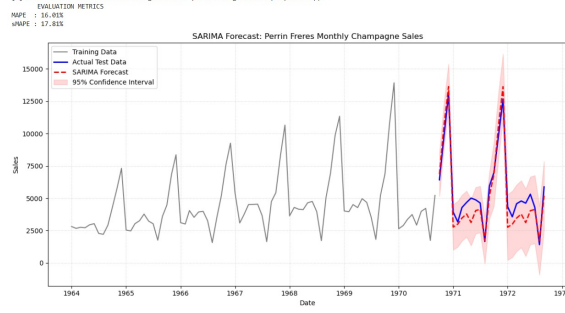


Figure 1 – SRIMA performance on Champagne dataset

4. RESULTS ON THESE DATASETS

Champagne Sales — SARIMA SARIMA was applied to the monthly champagne sales dataset and evaluated over a 24-month test horizon. As a purely univariate model with no access to external predictors, its performance relies entirely on the patterns learnable from the historical sales series itself. The model performs well on this dataset due to the strong, regular annual seasonality in champagne sales, which SARIMA captures through its seasonal differencing and seasonal MA term. Forecast accuracy degrades modestly at longer horizons (12+ months ahead), consistent with the theoretical property that SARIMA forecasts revert toward the series mean as the horizon grows. Evaluation is reported using MAPE and sMAPE, both percentage-based metrics that are scale-independent and appropriate for comparing across datasets: ??.

Tabela 2 – Final scores for SRIMA model on Champagne sales dataset

Model	Dataset	Test Horizon	MAPE (%)	sMAPE (%)
SARIMA(1,1,1)(0,1,1,12)	Chamooagne sales	24 months	16.01%	17.81%

For the second dataset,

Both ARIMAX and LSTM were evaluated on the same 168-hour (7-day) test horizon from the bike sharing dataset.

Clearly we can see that LSTM nearly halves the MAE and RMSE scores, ARIMAX’s forecast degrades sharply on the bike-sharing series because reliably estimating a seasonal component at period 24 is computationally prohibitive at scale, forcing a drastically reduced training window (7 days) that gives the model too little history to learn the daily demand cycle. The LSTM, by contrast, encodes

MODEL	TEST HORIZON	MAE	RMSE
ARIMAX	168 hrs	64.06 bikes	74.4 bikes
LSTM	168hrs	28 bikes	41.99 bikes

Tabela 3 – Caption

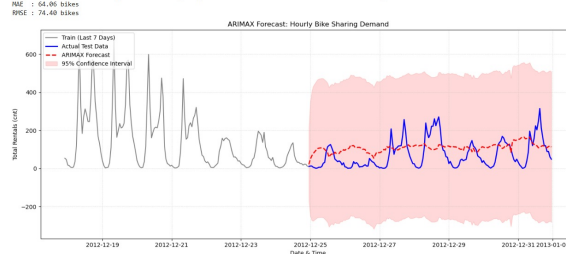


Figure 2 – ARIMAX Prediction

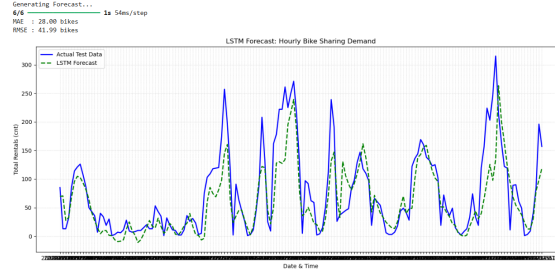


Figure 3 – LSTM prediction

the 24-hour pattern directly through its 24-step lookback window and learns it from a much larger training history, without needing an explicit seasonal parameter.

Here we saw a decent example of how complex and dense network such as LSTM was able to learn the non-linear relationship between current and historical data, where as classical model made more mistakes.

But deep learning models does not always outperform classical models. Using both of the paradigms together is the idea for the next section

5. HYBRID MODELS

The comparative analysis presented in the previous chapter demonstrates that neither classical statistical models nor deep learning architectures consistently outperform the other across all forecasting scenarios. Statistical models such as SARIMA and ARIMAX provide interpretable forecasts and perform well when the underlying data exhibits stable linear trends and seasonal patterns. Conversely, deep learning models such as LSTMs excel at learning nonlinear temporal dependencies and complex relationships from large multivariate datasets. These complementary strengths have motivated the development of *hybrid forecasting models*, which combine the interpretability of statistical approaches with the representational power of deep learning.

Hybrid forecasting models attempt to decompose the forecasting task into linear and nonlinear components. A statistical model first captures the deterministic linear structure of the time series, while a neural network learns the remaining nonlinear residuals that cannot be explained by the statistical model. The final

prediction is obtained by combining the outputs of both models, allowing each architecture to focus on the patterns it models most effectively.

5.1. MOTIVATION

Neither statistical nor deep learning models are universally optimal.

• Classical models require assumptions such as stationarity and linearity, making them less effective for highly dynamic datasets. Deep learning models, while capable of capturing nonlinear behaviour, generally require substantially larger datasets, greater computational resources, and often lack interpretability.

• Hybrid architectures seek to overcome these limitations by leveraging the strengths of both paradigms simultaneously. Such models have demonstrated improved forecasting accuracy across applications including electricity demand forecasting, traffic prediction, financial markets, and retail demand estimation.

5.2. COMMON HYBRID ARCHITECTURES

Several hybrid approaches have been proposed in recent literature.

- **ARIMA–LSTM:** ARIMA first models the linear trend and seasonality, while the LSTM predicts the remaining residual sequence.
- **SARIMA–LSTM:** Similar to ARIMA–LSTM but incorporates seasonal components explicitly before residual modelling.
- **CNN–LSTM:** Convolutional layers extract local temporal patterns, which are subsequently processed by LSTM layers to capture long-term dependencies.
- **Attention–LSTM:** An attention mechanism assigns greater importance to relevant historical observations before prediction.
- **TCN–Transformer:** Temporal Convolutional Networks capture local dependencies while Transformer attention models long-range temporal interactions.

These architectures consistently demonstrate lower forecasting errors than their individual components on complex multivariate datasets.

5.3. RESIDUAL LEARNING FRAMEWORK

Among existing hybrid strategies, residual learning has emerged as one of the most effective and intuitive approaches.

Let the observed time series be represented as

$$y_t = L_t + N_t,$$

where L_t denotes the linear component and N_t represents the nonlinear residual component.

A statistical forecasting model first estimates

$$\hat{L}_t.$$

The residual sequence is then computed as

$$r_t = y_t - \hat{L}_t.$$

An LSTM network is subsequently trained on the residual sequence to estimate

$$\hat{N}_t.$$

The final forecast becomes

$$\hat{y}_t = \hat{L}_t + \hat{N}_t.$$

This decomposition allows each model to specialize in modelling the characteristics for which it is best suited.

5.4. FUTURE SCOPE

The hybrid paradigm represents an active area of research. Recent work extends classical residual-learning frameworks by integrating Transformer architectures, graph neural networks, diffusion models, and foundation models for time series. Future forecasting systems are expected to combine statistical interpretability with large-scale pretrained models capable of generalizing across multiple forecasting domains while maintaining computational efficiency and robust uncertainty estimation.